IMPLEMENTATION BLUEPRINT

Index VWAP + Opening Range + 1M Execution

NIFTY | BANKNIFTY | SENSEX

Complete end-to-end phased delivery plan

Prepared 16 October 2026 | Strategy ID: IndexVwapOrb

SAME USER JOURNEY

Existing broker connection, account selection, option configuration, Start/Stop, monitoring and trade history. The new analyzer evaluates the selected index; existing execution services trade its CE/PE.

PLANNING STATUS

This document specifies proposed changes. No strategy was enabled and no trading implementation or live order validation has been performed. Unresolved rule choices are called out before implementation.

Readiness gates: valid VWAP input per index, locked retest/pivot rules, durable daily admission, and correct separation of index trigger levels from option premium risk.

Document guide

The blueprint follows the existing repository workflow and the corrected multi-index scope. Phase gates are acceptance conditions, not claims of completed work.

Scope and strategy rules

Supported indices, exchange routing, preserved account flow and exact entry/exit interpretation.

Current architecture and decisions

Source-backed integration points, data feasibility and rule choices that must be settled.

Phases 0-7

Rule lock, market data, analyzer, admission, execution risk, UI, replay and rollout.

Engineering appendices

Proposed data/API contracts, failure behavior, acceptance matrix and delivery responsibilities.

Evidence and release checklist

Original dashboard capture, chart limitations, readiness checklist and references.

Key operating assumption

One selected underlying per strategy instance, matching existing strategy pages. Two daily entries are shared across underlying changes for this account/strategy. This is a proposed baseline to lock in Phase 0; simultaneous independent instances and per-index trade quotas require an explicit policy change.

Scope and operating model

Scope correction: this strategy supports NIFTY, BANKNIFTY and SENSEX with the same account connection and option configuration flow as the existing strategies. NIFTY is only the illustration, not a restriction.

Selected underlying	Spot signal exchange	CE/PE execution exchange
NIFTY	NSE	NFO
BANKNIFTY	NSE	NFO
SENSEX	BSE	BFO, following contractManager's resolved instrument

Use instrument-master contract metadata as authoritative; do not copy inconsistent legacy UI exchange or lot-size constants. Broker connection is shared through the existing account/session provider; no separate strategy login or new credentials form. Preserve connection modal/status, reconnect handling and selected-account isolation.

Baseline operation matches existing pages: choose one underlying for a running strategy instance. Changing underlying while running must not silently replace the held contract or risk reference. Stop new scans and reconcile exposure before applying a new underlying; keep held-position monitoring tied to its immutable snapshot. Pending setups/overlays are scoped by underlying. Daily maximum remains two entries total per account/strategy/IST date across all three choices, so switching indices cannot reset the cap. Simultaneous multi-index operation is outside this baseline and would need separate instance IDs plus an explicitly shared admission policy.

1. Strategy interpretation

This is a selectable NIFTY, BANKNIFTY or SENSEX underlying signal strategy that buys CE for a bullish setup and PE for a bearish setup. It is not the existing option-premium range breakout and not the VWAP + SMMA placeholder.

1. Build opening range from the three completed 5M bars starting 09:15, 09:20 and 09:25 IST. Freeze highest wick and lowest wick at 09:30. Use the half-open interval [09:15,09:30); never include the 09:30-start bar in the range.

2. Wait for a subsequent completed 5M bar to close strictly above OR high or strictly below OR low. No entry on breakout alone.

3. Wait for a later retest/rejection. LONG: low <= OR high + tolerance and close >= OR high. SHORT: high >= OR low - tolerance and close <= OR low. No candle-colour filter.

4. Confirm a 1M swing with two candles on each side. A pivot is usable only once both right-side bars have closed. A later completed 1M close must break the confirmed swing high for LONG or swing low for SHORT.

5. Require entry close above session VWAP for LONG or below it for SHORT. Equality fails the VWAP filter.

6. Entry reference is that completed 1M close. LONG stop = retest low - 2 points of the selected index; SHORT stop = retest high + 2. Require finite positive risk. LONG target = entry + 2R; SHORT target = entry - 2R.

7. Accept at most two entries per IST trading day, one position for this strategy/account at a time. Stop new admission after 11:30. Maintain existing exposure through the configured global square-off time.

8. No strategy-generated discretionary exit. Generic position management owns SL, TP, manual exit and EOD. Contract resolver owns strike, expiry, lot size and quantity.

LONG example: OR high 25,150; later retest low 25,140 and close 25,155; confirmed 1M swing high 25,165; later 1M close 25,170 above VWAP. Stop 25,138; risk 32; target 25,234. These are NIFTY levels, never CE premium prices. SHORT mirror: OR low 25,000; retest high 25,010, close 24,995; swing low 24,985; entry 24,980 below VWAP; stop 25,012; risk 32; target 24,916. Examples are illustrative, not historical trades.

2. Current workflow and implementation evidence

Layer	Current source	What it does / implication
Application	README.md	React/Vite -> Node/Express/Mongo -> Python/FastAPI -> Angel SmartAPI
User journey	algo-trading/src/router/AppRouter.tsx; pages/StrategiesPage/StrategiesPage.tsx	Protected dashboard, strategy cards, individual pages, dynamic plug-in route, Trades and Search
Familiar strategy controls	algo-trading/src/pages/StrategiesPage/FiveMinBreakoutPage.tsx	Account-keyed settings, validated save queue, broker checks, Start/Stop, checkpoints, polling and mobile controls; use as UX reference
Generic plug-in page	algo-trading/src/pages/StrategiesPage/DynamicStrategyPage.tsx	Discovers manifests, but initializes SENSEX/quantity 20 and lacks equivalent account settings lifecycle; not ready for this strategy unchanged
Discovery	backend/services/strategyRegistry.js	Loads plug-in classes at startup; exact engineVersion 1.0.0; seeds missing per-user Strategy rows for all existing users
Interface	backend/trading/strategies/BaseStrategy.js	Single-timeframe manifest examples and option-centric analyze context; no enforced multi-stream schema
Validation/settings	backend/utils/inputValidation.js; services/strategySettingsService.js; models/StrategySettings.js	Strategy allowlist currently recognizes built-ins; settings scoped by user/account/strategy with optimistic versioning
Runner	backend/services/strategyEngine.js	Resolves contracts, scans CE/PE candles, constructs plug-in instances during scans; candle listener filters option token, exchange, interval and role
Feed	backend/services/marketDataService.js	Account/session-isolated feeds, interval candle builders, REST fallback and candle-closed events; foundation for spot subscriptions
Spot history	backend/services/candleService.js	NIFTY/BANKNIFTY NSE and SENSEX BSE index token mappings already exist; output strips volume; today's history cache can last five minutes, unsuitable as sole fresh 1M decision source
Orders/risk	backend/services/positionExecution.js; executionGateway.js; executionLeaseService.js	Durable intent, ownership, fill and recovery infrastructure. Held snapshot currently uses premium-based stops/targets and strategy-name branches
Charts	algo-trading/src/components/chart/TradingViewChart.tsx	Regular candles, markers, price lines, EMA/JMA series; extend with named VWAP series and time-bounded strategy overlays
Session clock	backend/services/marketSessionManager.js	Market-hours events and weekday checks; market close notification alone is not confirmed pre-close liquidation

Current user flow: sign in -> connect/select broker account -> choose strategy -> save configuration -> Start -> Node runner resolves CE/PE -> account market data -> plug-in evaluation -> execution admission/lease -> durable order intent -> Python broker order -> fill reconciliation -> position protection/monitor -> exit -> Trades/dashboard overlays.

Preserved new flow: same account connection -> same strategy card layout -> Index VWAP ORB page with familiar settings and controls -> Start -> spot 1M/5M evaluation -> existing CE/PE resolver and execution gateway -> same monitor, position status and trade history. Analysis moves to spot; user navigation and execution ownership remain familiar.

3. Decisions to settle before live implementation

The pasted text references an analyzer, but no NiftyVwapOrb analyzer was supplied or found in the strategy files inspected. Treat that text as requirements, not proof of implemented behavior.

A. VWAP volume source - blocking correctness decision

Session VWAP requires an explicit price/volume series. An index value is not itself a directly traded instrument; verify what the selected broker feed actually supplies for each of NIFTY, BANKNIFTY and SENSEX index volume. Do not invent volume, substitute equal weights, forward-fill it, or call a simple average VWAP. Existing chart history formatting omits volume regardless of upstream availability.

Phase 0 must capture sample broker OHLCV and document source, meaning, reset and availability. Proposed formula, subject to specification approval: sum(((H+L+C)/3)*barVolume)/sum(barVolume), from session-open completed 1M bars only. Aggregate cumulative tick volume into increments correctly. Missing/zero/invalid volume or incomplete session history blocks this strategy. A futures/ETF proxy is a strategy change, requiring an explicit choice and chart labelling, not a hidden fallback.

External verification: official Angel One SmartAPI documentation, checked 16 October 2026, defines historical records as timestamp/OHLC/volume. Its NIFTY-token example has volume zero. This supports making volume validation a feasibility gate; it does not prove all NIFTY candles always have zero volume.

B. Underlying risk versus option protection - blocking engine decision

Do not submit 25,138 as a CE stop price or compare PE premium to 24,916. Extend generic risk metadata with reference instrument and direction. Monitor spot thresholds: LONG exit when spot <= stop or >= target; SHORT exit when spot >= stop or <= target. Sell the held CE/PE through existing exit-intent execution. Persist spot signal entry separately from actual option fill and option P&L.

An underlying threshold is a software trigger, not a broker-native option stop. A separately configured premium emergency stop may provide broker-side protection, but requires a deliberate account risk policy; no guessed conversion via delta. On loss of fresh spot data, block entries, flag degraded protection, reconcile orders and apply the agreed generic outage policy. Underlying-index 2R is not a guaranteed option-premium 2R or realized currency return.

C. Ambiguous strategy details - proposed baseline, requiring rule lock

- Retest timeframe: use a later completed 5M candle, followed by 1M confirmation. The pasted specification does not explicitly fix retest timeframe.

- Tolerance: numeric index-point value still missing; validate separately for each underlying rather than assuming one tolerance fits all three. Expose a bounded validated field and do not invent a live default.

- Retest penetration: supplied inequality permits deep excursions before recovery. Preserve it unless a maximum penetration rule is explicitly added.

- Pivot selection: latest confirmed pivot formed after the retest; entry bar must be later than pivot confirmation. Define equal-high/low ties consistently. An alternative using a pivot formed before retest changes opportunities and must be chosen explicitly.

- Retest anchor: use the accepted retest candle extremum, freeze it for that setup. Specify replacement, expiration and invalidation rules before coding; do not add automatic VWAP or structure exits.

- New opposite breakout: proposed invalidation of the pending setup and creation of a fresh opposite setup; open position still exits only through generic risk management.

- Boundary: allow a bar whose close time is exactly 11:30 only if order admission occurs by 11:30:00 IST; a delayed event cannot submit later. Explicitly define timestamps as bar-open plus duration to avoid accidentally entering on an

11:30-start candle.

- Trade count: scope = user + broker account + strategy + IST date. Count one entry when any quantity first fills; additional fills of the same intent do not count again. Reserve capacity atomically before submission; unresolved submissions retain their reservation. Confirmed unfilled rejection/cancellation releases it. Partial fills count as one entry.

- One-position rule is per strategy/account, matching existing execution scope. Any account-wide cross-strategy restriction is a separate global policy.

- Global EOD time is not specified in the request and a dedicated square-off scheduler was not established by the inspected session code. Audit existing operations; implement a generic durable scheduler if absent. Do not confuse market-close events with successful liquidation.

Phase 0 - lock rules and establish baseline

Deliver a versioned rule contract resolving section 3; record broker candle timestamp/volume semantics; record current account settings, execution paths and existing checks without copying credentials. Verify holidays and session schedule behavior. Obtain at least one real completed morning session for each of NIFTY, BANKNIFTY and SENSEX with full 1M OHLCV and corresponding option data for validation.

Phase acceptance gate

VWAP source is valid and documented; retest/pivot/tolerance/cutoff/EOD rules agreed; original account UI screenshot retained; baseline checks recorded. Do not activate live mode while these remain unresolved.

Phase 1 - generic underlying multi-timeframe data

Extend BaseStrategy requirements compatibly with instrument role and multiple timeframes, e.g. UNDERLYING_CANDLES at ONE_MINUTE and FIVE_MINUTE. Keep existing manifests functional. Subscribe the selected NIFTY, BANKNIFTY or SENSEX spot instrument through the account feed; derive aligned 5M OHLCV from complete 1M bars or validate consistency with broker 5M bars. Backfill from 09:15 for late starts; do not use previous-day bars in OR/VWAP. Preserve OHLCV, source, close timestamp, completeness and freshness. Avoid duplicate history/live overlap and reject gaps or out-of-order input until repaired. Adapt runner event routing to admit underlying token/exchange events without weakening user/account/session filters.

Files: BaseStrategy.js, marketDataService.js, candleService.js/candleCache.js, strategyEngine.js and Python candle normalization if upstream drops volume.

Phase acceptance gate

exact three-bar OR at 09:30; no partial bars; no duplicate volume; same session results after late start; stale data blocks entries; existing option-only strategies behave as before.

Phase 2 - pure analyzer and plug-in adapter

Add backend/trading/strategies/indexVwapOrb.js and plugins/indexVwapOrbPlugin.js with ID IndexVwapOrb. Analyzer accepts immutable completed candle histories, explicit as-of time, locked rules and optionally serialized setup state; produces a deterministic result with no broker calls. Do not depend on long-lived plug-in instance memory because the current scan loop creates instances repeatedly.

Lifecycle: BUILDING_OR -> WAITING_BREAKOUT -> WAITING_RETEST -> WAITING_1M_CONFIRMATION -> SIGNAL_READY. Runner adds admission/exposure states. Preserve event ordering and pivot knowledge time; never identify a pivot using bars beyond evaluation time. Signal contains direction, reference instrument, signal bar close time, setup ID, OR levels, VWAP/source, retest/pivot timestamps, entry reference, SL, TP, risk points, rule version and failed reasons; isExit stays false.

Phase acceptance gate

deterministic fixtures for both sides, breakout without retest never enters, retest on wrong side never enters, two-right-bar pivot delay enforced, exact 2R and positive risk verified, no fabricated candle-colour filter.

Phase 3 - durable session admission and account registration

Add IndexVwapOrb to validated configuration with underlying enum NIFTY/BANKNIFTY/SENSEX; fixed 2R, swing length 2 and stop buffer 2 are displayed as locked rules. Keep execution quantity/expiry/strike policy in their existing layer. Introduce a durable session record keyed by user/account/strategy/IST date with unique index, setup checkpoint, processed candle/signal keys and entry reservations/counters. Integrate atomic reservation with existing leases/order intents; a non-transactional counter increment followed by an order submit is insufficient. Restore count and pending intents on restart before resuming entries. Session reset must not discard unresolved exposure.

Use registry's idempotent missing-record sync, but account for its all-users behavior: gate this plug-in to the intended account for rollout. Do not bulk enable strategies, overwrite old settings or consent, or reset accounts. Ensure new-user provisioning supports the plug-in. Account/session changes clear UI scope and reject stale callbacks.

Phase acceptance gate

third daily entry blocked under concurrent scans, reconnect and process restart; duplicate events cannot create another entry; uncertain submit cannot free capacity; wrong account cannot read or mutate settings/status.

Phase 4 - generic reference-based risk and execution

Extend Position/Trade/order-intent snapshots with risk reference type, reference token, direction, spot entry/stop/target, data timestamps and rule version. Preserve premium fill prices and P&L. Map LONG to buy CE and SHORT to buy PE via contractManager; reuse instrument-master lot/expiry metadata and existing liquidity/margin/quote checks. Do not hardcode lot size from existing UI constants.

Extend positionExecution and generic monitoring to evaluate spot risk, retain premium emergency protection only under agreed policy, and persist enough metadata for full restart recovery. Retain underlying subscriptions while exposed after 11:30 and after scanner Stop. Add/reuse a global EOD scheduler with reconciliation, partial-fill handling and idempotent exit intents. Show exit pending until broker confirmation; stop scanning must not falsely show a held position as closed.

Phase acceptance gate

CE/PE exits trigger on the correct spot inequality, option premiums never mixed with index points, concurrent target/EOD/manual events produce one sell intent, partial fills and recovery preserve risk, lost data is visible and handled.

Phase 5 - same UI journey and chart overlays

Add a card/sidebar entry and /strategies/index-vwap-orb page using existing StrategiesLayout, account settings lifecycle, broker checks, live consent, responsive controls and polling conventions. Prefer a dedicated page matching FiveMinBreakoutPage until DynamicStrategyPage has equivalent scope/settings behavior. Prevent duplicate card discovery if the page is explicitly registered.

Keep the familiar layout: header/actions -> settings -> live checkpoints -> position summary/logs. Settings include the same NIFTY/BANKNIFTY/SENSEX selector, broker connection/account controls, strike mode ATM/ITM/OTM and depth, premium range controls where used by the existing resolver, expiry handling, quantity and live-trading consent as the existing strategies. Preserve existing expiry behavior; if it is automatic today, do not invent a new mandatory expiry-selection step. Show the approved underlying-specific tolerance. Show OR window, entry cutoff, pivot length, buffer, daily cap and 2R as read-only rules. Do not offer irrelevant premium-range lookback, discretionary candle exits, or minimum RR.

Monitor checkpoints: broker connected; spot 1M/5M ready; volume/VWAP ready; OR locked; breakout closed; retest held; pivot confirmed; entry admissible; contracts ready. Display entries used/reserved out of two and precise waiting/blocking reason. Keep monitoring state distinct from execution states such as POSITION_UNPROTECTED, EXIT_PENDING and RECONCILIATION_REQUIRED.

Chart retains current toolbar, normal candles, timeframe/date controls and overlay toggles. Add OR high/low from 09:30 onward, shaded opening window, named VWAP line, breakout/retest/pivot/entry markers, spot SL/target after entry and 11:30 cutoff. Use the same backend diagnostics for chart and decisions. Mark 5M events at their close/knowledge time on a 1M chart; show pivot formation and confirmation separately. Historical overlays require a completed replay, never imply a real order fill. In position/trade cards distinguish selected-index trigger levels from option entry/premium/quantity/currency P&L.

Phase acceptance gate

desktop/mobile visual comparison preserves workflow; account switching never leaks settings, counters or overlays; Stop/Exit behavior matches existing protected-position semantics; refresh reloads saved validated configuration and status.

Verify all three underlying selections, NSE/NFO versus BSE/BFO routing, resolver expiry/strike/lot metadata, premium filters, broker disconnect/reconnect, underlying changes, and account switching. Volume feasibility is evaluated per index; valid data for one does not enable another with missing volume. Apply replay, admission, risk and rollout gates to all three supported indices.

Phase 6 - replay and paper acceptance

Replay completed sessions sequentially through the same analyzer and runner gates. Include no-breakout, no-retest, failed retest, failed VWAP, simultaneous timeframe close, both sides, missing opening bars, zero volume, duplicate/out-of-order events, late start, cutoff boundary, restart after two fills, broker rejection/timeout, partial fill and EOD recovery. Paper fills must use option quotes with timestamps and explicit assumptions; spot points cannot stand in for premium P&L.

Run relevant existing backend execution/account/feed/recovery suites and frontend settings/account-isolation suites, plus new meaningful analyzer/session/risk tests. Verify UI overlays against the decision trace. Report sample coverage and observed behavior; a clean paper run does not establish profitability.

Phase acceptance gate

no unexpected entries, exact rule trace for every entry, no unresolved exposure, restart/counter checks pass and existing strategies regress cleanly.

Phase 7 - account-scoped rollout and operations

Deploy behind an account/strategy feature gate initially stopped, with live consent off. Reconcile existing exposure first. Start with shadow evaluation, then paper operation, then user-controlled live activation after readiness review. Monitor stale data, volume validity, blocked reasons, daily reservations/fills, signal-to-submit delay, protective status, exit acknowledgements and EOD completion. Roll back by blocking new admissions while preserving held-position monitoring and reconciliation.

Phase acceptance gate

target account can see/save/start/stop using the established flow; no other account is silently activated; operational runbook and rollback are verified. Scheduling estimates should follow Phase 0 because feed validity and risk-engine scope determine most effort.

5. Visual evidence and limitation

Original capture: output/nifty-vwap-orb/original-dashboard.jpg, taken from the running localhost dashboard on 16 October 2026. The user is signed in, but broker connection is offline and the NIFTY chart is empty. No authentic candle screenshot or historical setup could be verified from that screen. The accompanying overlay is explicitly illustrative and uses the numeric LONG example above. Replace it with a verified completed-session chart annotation during Phase 0 once valid chart data is available.

Illustration: output/nifty-vwap-orb/illustrative-strategy-overlay.png. This generated image is a conceptual storyboard, not a plot to scale: its drawn candles/right-hand axis are not numerically consistent with the callouts, and its pivot drawing does not independently prove the 2+2 confirmation. Use the written numeric example and rule sequence as authoritative. It must not be used to validate a historical signal or fill. A genuine annotated chart remains pending valid market data.

6. Recommended implementation order

Rules/data feasibility -> closed spot streams -> deterministic analyzer -> durable admission -> reference-based position risk/EOD -> familiar UI/overlays -> replay/paper -> account-scoped activation. Do not treat adding a manifest/card as a working trading integration: current validation, runner routing and risk snapshots all require changes.

Appendix A - Engineering contracts

Configuration and settings

Proposed fields: strategyId, underlying, strikeMode, strikeDepth, quantity, premiumMin/premiumMax where supported, approved retestTolerancePoints and liveTradingConsent. Reuse automatic expiry policy unless the existing account flow exposes a selector. Validate quantity against resolved lot metadata on both save and start. Fixed rules are server-authoritative: riskReward=2, swingLength=2, stopBufferPoints=2, entry cutoff 11:30 IST and daily cap 2. The broker account comes from verified account context, never a trusted arbitrary request field.

Analysis input

Use completed 1M/5M OHLCV, underlying identity, session date, asOf timestamp, volume-source descriptor, rule version and setup checkpoint. Each bar carries openTime, closeTime, completeness and source. Freshness and session completeness are explicit; the analyzer receives no credentials and performs no broker mutation.

Analysis output

Use NONE/ENTRY with direction LONG/SHORT, setupId, signalId, signalClosedAt, referenceInstrument, entryReference, stopReference, targetReference, riskPoints and isExit=false. Diagnostics carry OR levels, VWAP, retest candle, pivot formation/confirmation timestamps and blocking reasons. Map this result through an adapter to the current runner interface; do not assume the existing option-centric result shape can carry it unchanged.

Session persistence

Proposed StrategySession collection: userId, brokerAccountId, strategyId, istDate, entriesFilled, reservations, ruleVersion, setupByUnderlying, lastProcessedByStream and version. Unique scope index protects one daily record. Signal identity includes underlying, direction, setup identity and closed entry-bar timestamp. Use atomic conditional admission under the existing execution lease; verify transaction support before selecting a cross-document transaction implementation.

Held-position snapshot

Persist underlying token/exchange, option contract token/exchange/symbol/expiry, actual filled quantity and premium average, index entry/stop/target, risk direction, rule version, execution mode and generic protection policy. Changes to saved settings must not rewrite an open position. Exit records retain trigger index price/time separately from actual option sell fills and realized P&L.

Backward compatibility

Add fields and capabilities without reinterpreting old premium-based positions. Missing riskReferenceType on an existing position continues the documented legacy behavior. Restore a new index-reference position only from sufficient durable metadata; otherwise show reconciliation required rather than infer levels from current settings.

Appendix B - UI and service integration

Connection and launch sequence

Dashboard -> existing Connect Angel One modal if needed -> selected broker account -> Strategies -> Index VWAP ORB -> select NIFTY/BANKNIFTY/SENSEX -> familiar option settings -> validated Save -> existing live consent if applicable -> Start. Display data readiness before admitting entries. Broker reconnection uses the current provider and session generation rules, not a second login flow.

Existing API family

Reuse GET /strategies/manifests, GET /strategies/IndexVwapOrb/status, POST /strategies/IndexVwapOrb/start, /stop and /exit after confirming route middleware. Reuse the existing settings API rather than introducing a separate persistence path. Add optional status diagnostics compatibly. Status reads, chart mounts and polling must not submit orders or enable the strategy.

Status model

Report scan stage separately from execution exposure. Include selected underlying, connection health, data freshness, OR/VWAP readiness, setup stage, entry blockers, entries filled/reserved, contract resolution and held-position state. Preserve POSITION_PROTECTED/POSITION_UNPROTECTED, EXIT_PENDING and RECONCILIATION_REQUIRED semantics. A software spot trigger alone must not imply working broker-native protection.

Chart behavior

Use regular selected-index candles for decisions and overlays. Keep the existing chart toolbar and user toggles. Display the opening window, frozen OR lines, session VWAP, retest/pivot knowledge markers and post-entry index SL/target. Option premium and fills remain in the position summary/history. UI rendering follows backend analysis diagnostics, avoiding a separate browser-side trading algorithm.

Stop, exit and selection changes

Stop disables new entries and setup scanning while held exposure remains managed. Exit invokes the existing coordinated exit path and stays pending until reconciliation. Underlying/account changes must preserve original position monitoring and reject stale UI responses. Do not silently clear daily usage on refresh, reconnect, index change or strategy restart.

Appendix C - Failure and recovery behavior

Incomplete opening range / invalid volume

Block entry and state the missing bars or volume source problem. Backfill and re-evaluate only using information available as of the event; do not create invented history.

Duplicate, delayed or out-of-order event

Deduplicate by stream and candle identity. Repair order and gaps before analysis. Admission checks the current cutoff as well as signal close time; late delivery cannot bypass the entry window.

Entry timeout or partial fill

Retain reservation and durable intent until broker reconciliation. Never submit a second entry merely because acknowledgement is missing. First non-zero fill consumes one daily entry; additional fills do not add another count.

Spot or option feed failure

Block new entries. Keep exposure visible, distinguish missing trigger data from missing execution quotes, and apply the approved generic outage policy. Do not claim a historical threshold fill after data returns.

Concurrent target, stop, manual or EOD trigger

Route all through the same coordinated exit intent. Reconcile protection and remaining quantity before additional sells. A stop-order fill racing a manual sell must not produce an unintended short option position.

Backend restart or session replacement

Restore leases, unresolved intents, daily reservations and held risk snapshots before admission. Recover subscriptions for both the index risk reference and held option. Reconstruction does not automatically enable new scanning.

Database unavailable

Block new entry if durable admission cannot be recorded. Preserve uncertain broker outcomes for recovery and alert through existing mechanisms; do not infer completion from a failed persistence write.

EOD exit incomplete

Persist exit pending, reconcile residual quantity and expose operational attention. A scheduled request or market-close event is not proof of square-off. Keep a documented manual broker recovery route.

Appendix D - Acceptance and regression matrix

Strategy semantics - all three indices

LONG and SHORT fixtures; exact opening-range interval; completed 5M breakout; later accepted retest; two-left/two-right confirmed pivot; later strict close break; VWAP side; finite risk; exact 2R; no unintended candle-colour filter.

Time and data boundaries

IST date reset, machine timezone independence, 09:30 range lock, close-time convention, exact 11:30 boundary and delayed event, session holidays, late-start backfill, missing bars, zero volume and duplicate history/live overlap.

Routing and option configuration

NIFTY/BANKNIFTY NSE spot to NFO options; SENSEX BSE spot to resolved BFO options; ATM/ITM/OTM depth, premium filters, expiry policy, lot-size validity, quantity, broker/account identity and stale contract rejection.

Admission concurrency

Two accepted entry fills and third blocked; reservation races; confirmed unfilled rejection releases capacity; unknown submission does not; partial fill counted once; index switch, reconnect and restart do not reset the cap.

Risk and recovery

Correct LONG/SHORT index exit inequalities; premium fills/P&L remain separate; restart preserves levels; Stop retains monitoring; concurrent triggers yield one coordinated sell; partial exit uses remaining quantity; EOD completion is reconciled.

UI compatibility

Desktop/mobile control parity, saved settings restore, account switch isolation, no duplicate discovery card, meaningful readiness messages, daily counter, chart toggle behavior, and separate index trigger versus premium labels.

Existing strategy regressions

Run relevant baseline strategy tests, account-isolation, execution gateway, position execution, feed isolation, startup recovery and frontend settings lifecycle suites. Compare existing signal/risk fixtures; no unintended behavior change is acceptable.

Evidence pack

For each case record input fixture/session, rule version, evaluation time, expected result, actual result, diagnostic trace and linked test output. Include verified real chart annotations for each supported index once valid data exists; label replay and paper executions accurately.

Appendix E - Delivery ownership and release

Phase ownership

Phase 0: strategy/product owner plus market-data engineer. Phase 1: feed/backend owner. Phase 2: strategy/backend owner. Phase 3: persistence/execution owner. Phase 4: execution and operations owners. Phase 5: frontend owner with backend diagnostics support. Phase 6: QA plus strategy/execution owners. Phase 7: release/operations owner and account operator. These are responsibilities, not assignments to named people.

Reviewable deliverables

Deliver phase changes as bounded reviews: (1) rule contract/data evidence, (2) compatible feed requirements and spot streams, (3) pure analyzer and fixtures, (4) settings/session admission, (5) generic risk and recovery, (6) familiar UI and overlays, (7) replay/paper evidence, (8) account rollout and runbook. Each review lists behavior, affected files, checks, migration and rollback implications.

Dependencies and estimation

Phase 0 gates the data model and analyzer. Phase 1 enables Phase 2. Phases 3 and 4 must integrate before executable entries. UI structure can be prepared after rule lock, but full acceptance depends on real status/risk data. Replay requires all execution and UI contracts. Estimate effort after data feasibility and generic risk scope are known; no unsupported delivery dates are committed.

Migration and rollout

Create additive fields/indexes, retain compatibility with existing positions, and register the new strategy idempotently. Gate account availability independently from registry discovery. Default stopped and live consent off. Roll out shadow evaluation, then paper, then operator-enabled live use after gates pass. Do not silently backfill consent or activate accounts.

Rollback

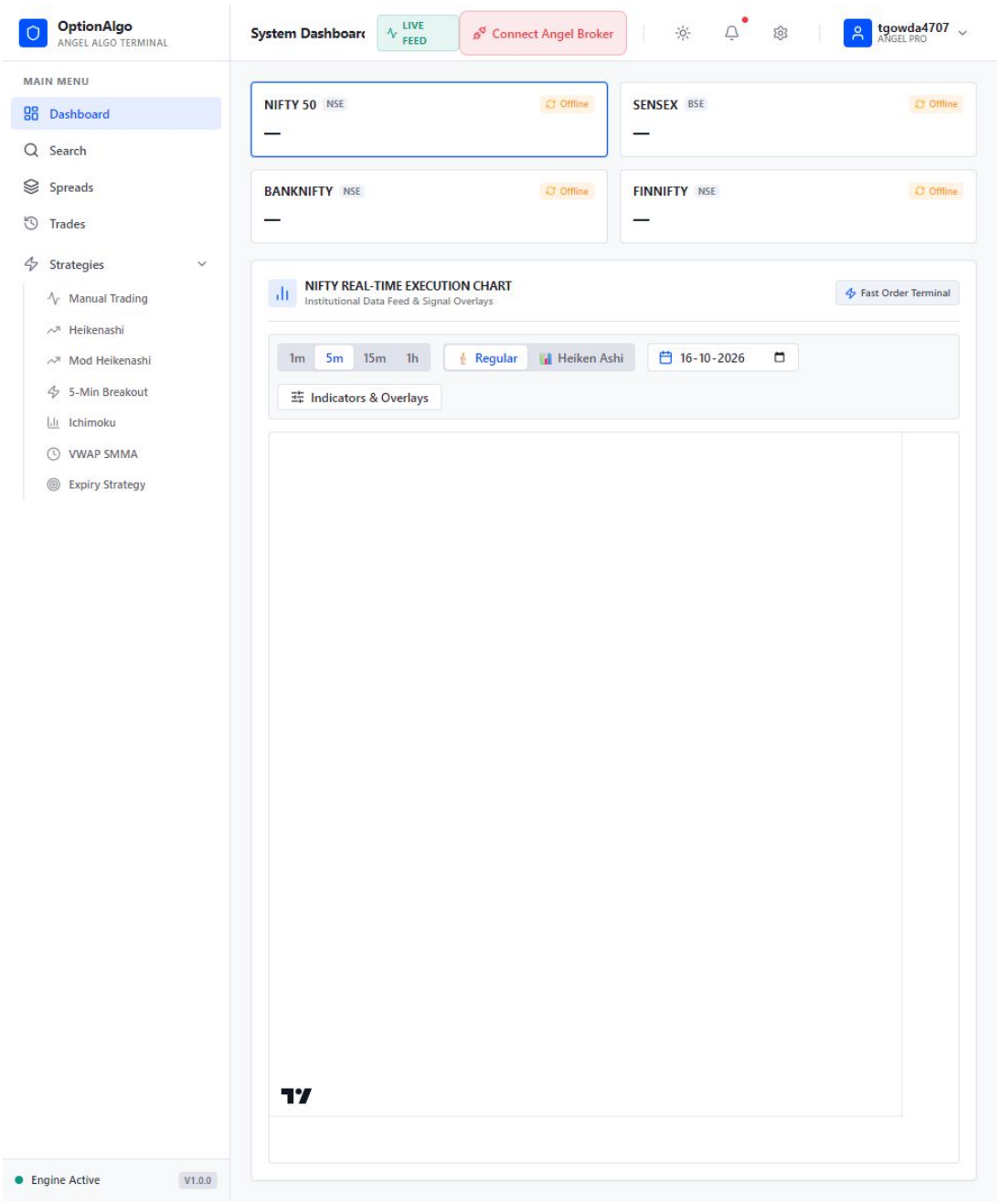
Disable new IndexVwapOrb admission first. Continue monitoring and reconciliation for held positions using a compatible release. Avoid reverting to code unable to understand index-reference risk while exposure remains. Preserve session counters, intents, fills and snapshots; remove feature visibility only after operational continuity is confirmed.

Operational checklist

Before session: broker/account health, feed freshness and volume validity, contract metadata, session date, daily reservations, existing exposure and global square-off schedule. During session: stage/blockers, delays, intents, partial fills and protection. After cutoff: no new admissions, held monitoring active. After square-off: verify zero remaining quantity or explicitly unresolved status, save outcomes and reconcile the daily entry ledger.

Appendix F - Original UI evidence

The untouched screenshot below was captured from the running local dashboard. Angel One was disconnected and the selected NIFTY chart was empty. It confirms the existing layout, not a real strategy setup. The previous AI-generated storyboard has known scale inconsistencies and is deliberately excluded as a validation chart.



Required follow-up evidence: valid completed-session candles and rule-derived overlays for NIFTY, BANKNIFTY and SENSEX.

Final readiness checklist and references

- Rules locked: tolerance per index, retest timeframe, pivot timing/ties, invalidation, cutoff and daily-count policy.
- Valid and documented VWAP input for each enabled index; no silent proxy or equal-weight substitute.
- Closed 1M/5M data, correct exchange routing, full-session backfill and freshness checks verified.
- Pure analyzer produces deterministic diagnostics without future-bar information.
- Daily admission survives races, ambiguous submissions, index changes and process restarts.
- Option contract metadata, margin/quote checks, actual fills and index-reference exits reconcile correctly.
- Global EOD behavior and outage/manual recovery policy are operationally verified.
- Same connection/settings/control flow works on desktop/mobile with correct account isolation.
- Replay, paper and existing-strategy regression evidence is reviewed for all three indices.
- Feature gate, stopped default, live consent, monitoring and rollback are ready for the target account.

Source basis

Repository review and corrected implementation plan: NIFTY_VWAP_ORB_IMPLEMENTATION_PLAN.md. User-provided strategy contract: attached Pasted text.txt. Key source modules are listed in section 2. Proposed modules, collections and APIs in appendices are design proposals; their existence is not implied.

External source

Angel One SmartAPI documentation, checked 16 October 2026. Historical records include timestamp/OHLC/volume; the documented NIFTY example uses zero volume. Actual broker samples must establish suitability for each enabled index.

Completion of this PDF is documentation delivery only. Phase gates, testing and live readiness remain implementation work.